AKSHAYA B

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PROFESSIONAL SUMMARY

AI/ML Engineer with 1+ year of production experience building LLM and anomaly detection systems for enterprise cybersecurity. At SISA Information Security, delivered a RAG pipeline over large-scale SIEM logs for SOC triage, UEBA anomaly detection models within the ProACT AI/ML engine spanning Windows, Linux, Data Exfiltration, and Reconnaissance log sources, and an AI-powered context aware DFIR chatbot for structured forensic incident response. Work spans the full stack applied ML, data engineering, LLM orchestration, and DevOps. Currently focused on imporvising intelligent threat detection and automated SOC operations.

WORK EXPERIENCE

SISA Information Security — Bangalore,IN

AI ML Engineer — Jul 2025 – Present

1.**DFIR Chatbot** Architecting and Developing an AI-powered Digital Forensics & Incident Response (DFIR) chatbot to guide investigators through structured response workflows, automate evidence collection guidance, and improve consistency in forensic investigations and repetitive reporting.
2.**LLM Orchestration —** Designing multi-turn conversation flows using LangChain with context-aware prompting to handle complex investigative queries across incident timelines
3.**UEBA Anomaly Detection** — Developed and deployed anomaly detection models within SISA's ProACT AI/ML engine, covering Windows, Linux, data exfiltration, and reconnaissance log sources to strengthen automated threat detection across large-scale security telemetry
4.**ML Pipeline Optimization** — Configured and optimized ML engines within a production SIEM platform at the intersection of DevOps and data engineering, improving reliability and intelligence of security data pipelines for SOC operations

AI/ML Intern — Jul 2024 – Jul 2025

1.RAG Pipeline — Designed and deployed a Retrieval-Augmented Generation (RAG) pipeline over large-scale SIEM telemetry, reducing LLM hallucinations and improving alert triage relevance for SOC analysts.
2.Vector Search — Processed and embedded millions of security log records into Pinecone,Milvus and Chroma vector databases, enabling low-latency semantic search across historical incident data.
3.API Development — Built production-grade FastAPI inference endpoints, integrated with an Angular frontend for real-time incident management; maintained CI/CD pipelines and model versioning across staging and production
4.Threat Intelligence — Contributed to an LLM-powered threat investigation system that reduced manual analyst triage time by surfacing contextually relevant historical alerts

EDUCATION

1.Dayananda Sagar University — Bangalore, IN
Bachelor of Engineering — June 2025
Major in Computer Science (Data Science)
Cumulative GPA: 9.52/10

TECHNICAL SKILLS

AI & ML: LLMs, RAG, LangChain, Hugging Face Transformers, Scikit-learn, TensorFlow, PyTorch, CNNs, NLP
Cybersecurity AI: Threat Intelligence, DFIR, SOC Alert Triage, SIEM/ELK Stack, Incident Response Automation
Databases & Vector Stores: Pinecone, Chroma, MySQL, Elasticsearch, Kafka
Backend & APIs: FastAPI, REST APIs, Streamlit, Uvicorn
DevOps: Git/GitHub, CI/CD, Docker, Model Versioning
Languages: Python, Java, HTML/CSS/JavaScript (basic), C (basic)
Visualization: Power BI, Matplotlib, Seaborn

PROJECTS

1.Skin Care Recommendation System | [GitHub]

Developed a skincare recommender system that analyzes user-provided skin images to offer personalized product suggestions. Used TensorFlow CNN to classify skin types (dry/oily) with 91.66% accuracy. Integrated a YOLOv8 model for detecting skin issues like blackheads, papules, and pustules. Leveraged a custom-trained Roboflow API for acne severity classification. Built a recommendation engine using cosine similarity and Count Vectorization. Deployed the solution with a Streamlit-based user interface.
Publication DOI: 10.1109/ICAECA63854.2025.11012227

2.California House Price Prediction | [GitHub]

It is an End to End Machine Learning Project built to predict the house prices in California.
The preprocessing tasks are performed in a pipeline that includes data cleaning,scaling,feature selection and Data Analysis.
The model utilized many models like **Decision Tree Regressor,Linear regression,Random forest regressor** to predict the prices using **RMSE** as an error analysis metric.Finally the model parameters were hyper tuned using GridSearchCV.

3. Compound Facial Emotion Classification System

Designed and implemented a multi-stage facial emotion recognition system that integrates machine learning, deep learning, and texture-based analysis to classify compound emotions with high accuracy. The system includes a robust preprocessing pipeline.Models such as Haar Cascade for face detection, KNN, SVM, CNN, VGGNet, and EfficientNet were trained for feature extraction and classification, while Local Binary Patterns (LBP), facial landmarks, and morphological analysis were incorporated to capture fine-grained expression details and temporal motion features. The system was evaluated using accuracy, precision, recall, and F1-score, demonstrating improved recognition of complex and subtle emotional states.

ACHIEVEMENTS AND INTERESTS

- **Rising Star Intern Award** December -SISA Information Security
- **SPOT Award** — Recognised for outstanding contributions in Q3, SISA Information Security (Feb 2026)
- **Tech Wizard Award** — Recognised for outstanding contributions in Q4, SISA Information Security (Apr 2026)
- Volunteering for charities and college events.
- Reading Books